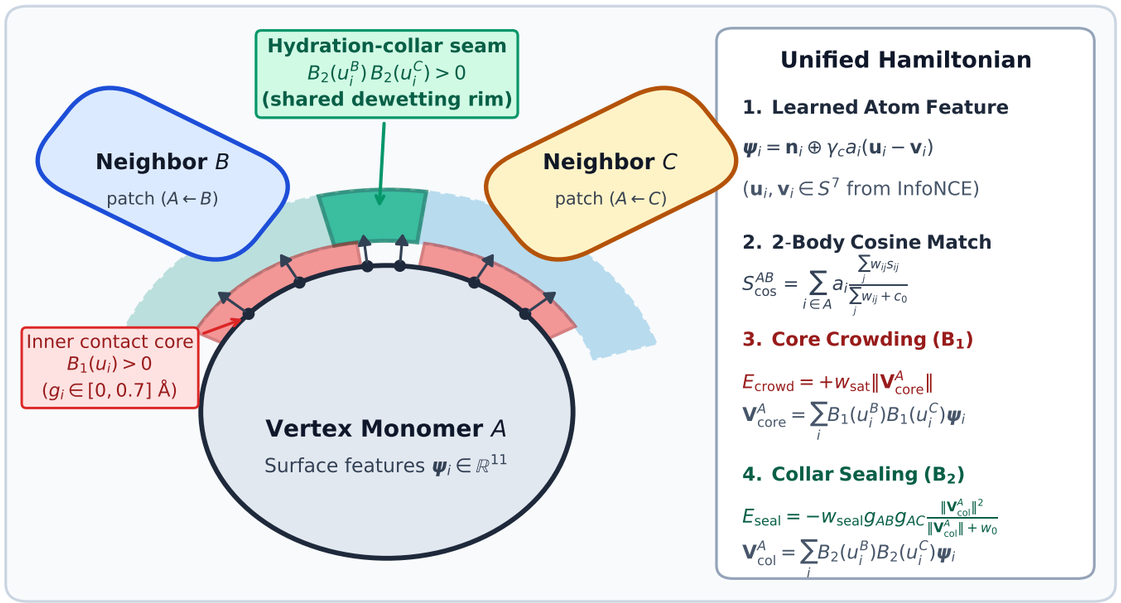
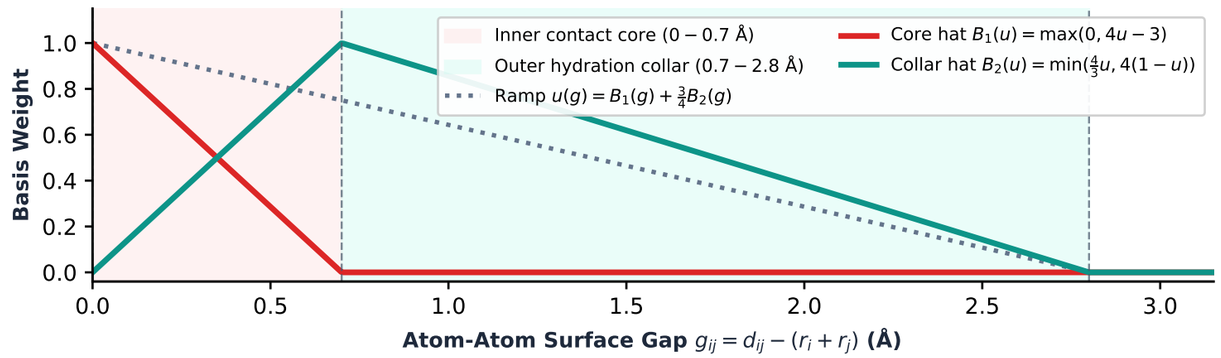
Cutoff-Free Piecewise-Linear Surface-Field Hamiltonian for 3D Protein Crystallization

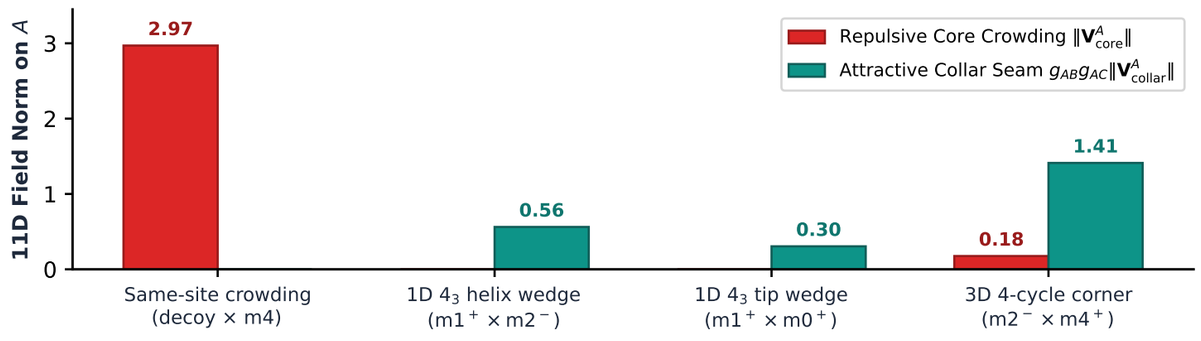
Unifying 2-body contrastive cosine embeddings $\boldsymbol{\psi}_i = (\mathbf{n}_i,\ \gamma_c a_i(\mathbf{u}_i - \mathbf{v}_i))$ with radial hat functions $B_1(g), B_2(g)$ eliminates $s > 0.5$ cutoffs and triangle rules

a Piecewise-Linear Radial Surface Field on Vertex Monomer A

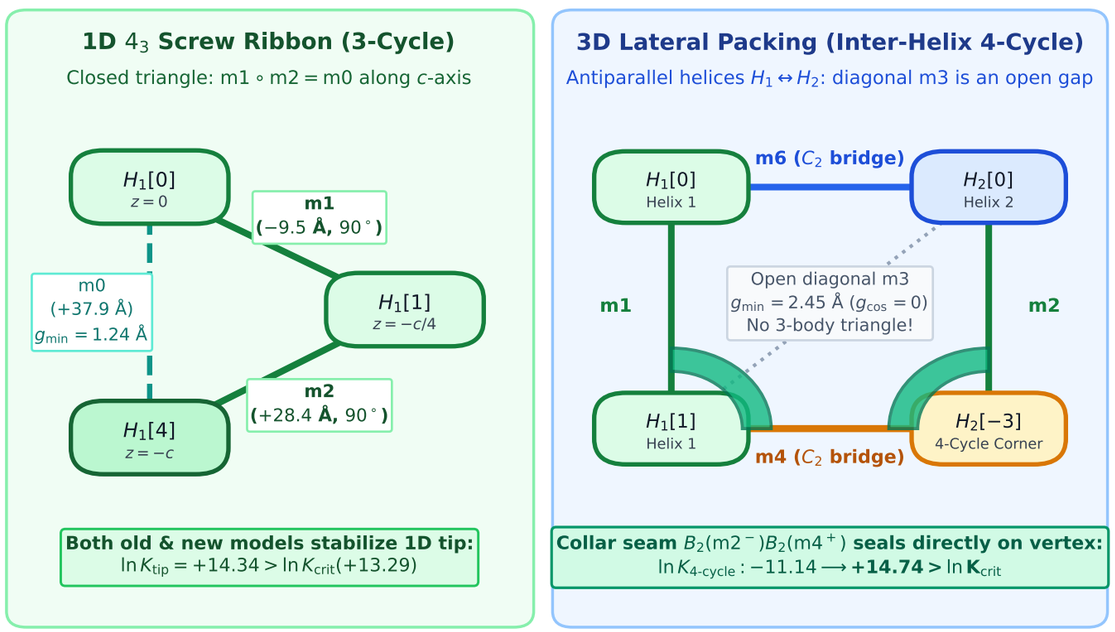


b 2-Knot Piecewise-Linear Radial Hat Basis & 1L2Y Vertex Overlaps





c Why Pure-Vertex Surface Fields Resolve Both 1D 3-Cycles & 3D 4-Cycles



d Multi-Body Crystal Growth Site Constants $\ln K_{\text{site}}$ (`pair-binding`)

